DS 5216 - Artificial Intelligence

Programming Assignment 02

Performance Comparison and Discussion Report

Name:M.U.K.Gunawardhna

Index No:DTS2426

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1. Performance Evaluation of the Player Tracking Model

1.1 Experimental Setup

To use a YOLO-based object detection model to detect and track players in sports videos, a dataset of clips was created using publicly available sources of sports footage (football, cricket, rugby matches) that were processed into frames for annotating where players could be found. The model was trained using the PyTorch framework and evaluated based on common object detection performance metrics (i.e., accuracy, precision, recall, F1-score, mean Average Precision (mAP)).

1.2 Performance Comparison

1.2.1 Player Detection Results

Metric	**Value**
Input Resolution	640 × 384
Number of Detected Players	2 Persons
Inference Time	1001.0 ms
Preprocessing Time	4.0 ms
Postprocessing Time	2.0 ms
Accuracy	1.000 (100%)

Table 1.Performance Metrics of Player Detection

As all 3,812 of the proposed instances were evaluated to have been classified correctly, there was only one class ("person") represented in both the actual and predicted labels of the evaluation dataset, which caused a warning from the sklearn library about the confusion matrix dimensions. However, the model's classification accuracy was perfect for the proposed evaluated samples.

1.2.2 Keypoint Detection Results

Metric	Value
Input Resolution	640 × 384
Number of Detected Players	3 Persons
Inference Time	151.2 ms
Preprocessing Time	3.7 ms
Postprocessing Time	1.3 ms
Accuracy	0.3827 (38.27%)

Table 2.Performance Metrics of KeyPont Detection

Confusion Matrix is as follows: 1,459 / 2,353 (True/False). One of the generated warning messages from sklearn was due to the dataset used for evaluating the performance of our keypoint detection model did not contain any negative samples meaning that computations for metrics like false positive rate were difficult to make.

1.3 Comparison of Model Performance

Performance Measure	Player Detection	Keypoint Detection
Accuracy	100.00%	38.27%
Preprocessing Time	4.0 ms	3.7 ms
Inference Time	1001.0 ms	151.2 ms
Postprocessing Time	2.0 ms	1.3 ms
Detection Capability	Excellent	Moderate

Table 3.Comparison of Model Performance

1.4 Discussion of Model Performance

The performance of the player detection model is outstanding due to the high accuracy rate achieved of one hundred percent when using a YOLO-based detection mechanism. In this type of model, detection of players in sports videos occurred very quickly, however, due to the limited amount of time taken for detection of any object (1001 milliseconds), it may not be possible to implement this type of model in a real-time application without using hardware acceleration.

In contrast to the player detection model, the keypoint detection model has a much lower accuracy at 38.27%. However, this model has a much faster inference time than the player detection model. In the keypoint detection model, the model's inability to reliably detect the body joints of players can be attributed to several reasons. These include occlusion of players, motion blur, camera angle differences, and the deficiencies present in the training dataset. Additionally, the evaluation warnings generated by the scikit-learn library indicate that the keypoint model had a test dataset that contained only positive examples, which limited the calculations of some classification metrics. So that a true measure of performance can be determined, future evaluations need to contain both positive and negative examples.

1.5 Limitations

- Presence of only one class in the evaluation dataset.
- Player occlusions during gameplay.
- Fast player movements causing detection errors.
- Limited dataset size.
- Variations in lighting conditions and camera angles.
- Computational limitations affecting model performance.

1.6 Possible Improvements

- Add additional training videos and annotations.
- Include negative samples in your evaluation.
- Implement more sophisticated tracking algorithms such as DeepSORT or ByteTrack.
- Utilize data augmentation techniques.
- Increase the number of epochs on which to train your model.
- Use GPUs to accelerate processing speeds.
- Use better architecture for pose estimation models to find keypoints.

1.7 Conclusion

The overarching aim of this system was to show how deep learning techniques can be effectively applied to sports analytics through a player tracking system that has been developed. The player detection model built using the YOLO algorithm was successful at achieving high accuracy with player detection, whereas the keypoint detection model had moderate performance and requires additional optimisation. Overall, this project has confirmed that computer vision-based player tracking technologies can be used effectively within sporting video analysis.